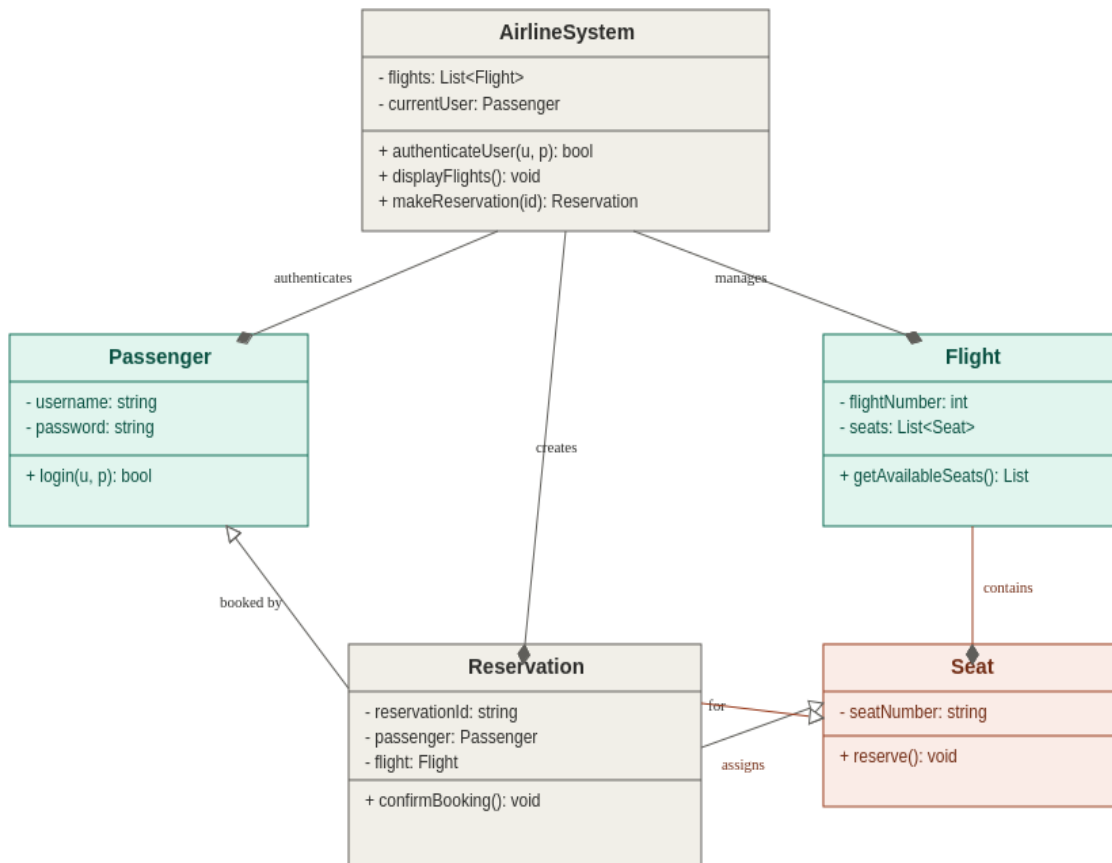


Class Diagram

The class diagram models the logical structure of the system as a set of collaborating objects: Airline System coordinates the workflow, Passenger and Flight hold the core data, Seat belongs to a Flight, and Reservation ties a passenger, flight, and seat together once a booking is made.

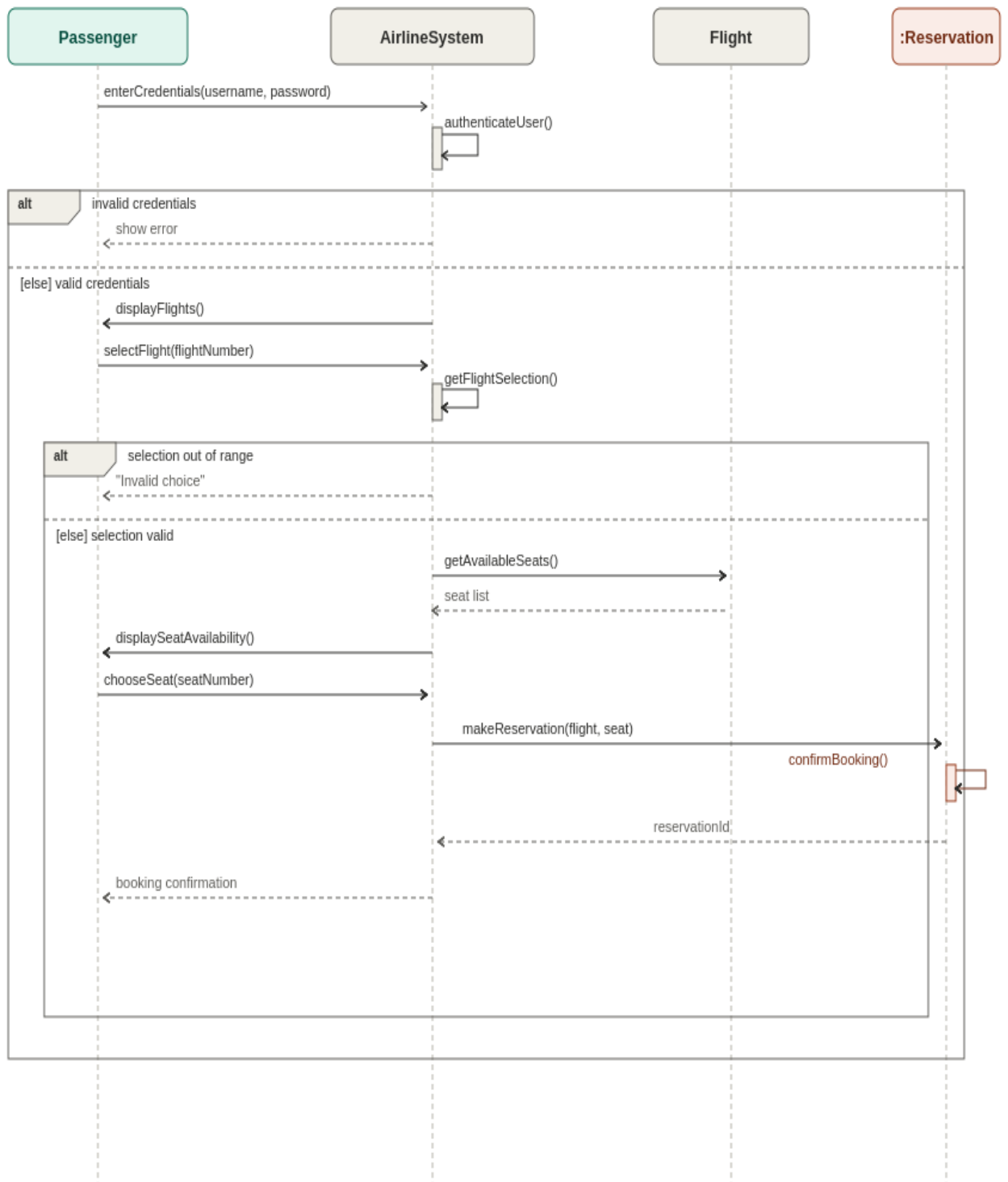


Class summary

Class	Responsibility
Airline System	Coordinates the whole workflow: authenticates the passenger, lists flights, and creates reservations.
Passenger	Holds login credentials and identifies who a reservation belongs to.
Flight	Represents a scheduled flight and owns its collection of seats.
Seat	Represents a single seat on a flight and whether it is available.
Reservation	Links a passenger, a flight, and a seat together and confirms the booking.

Sequence Diagram

The sequence diagram shows the interaction between a Passenger and the Airline System during a full booking, including the two failure paths (invalid login and out-of-range flight selection) drawn as alt frames, and the collaboration with Flight and Reservation once a valid flight is chosen.

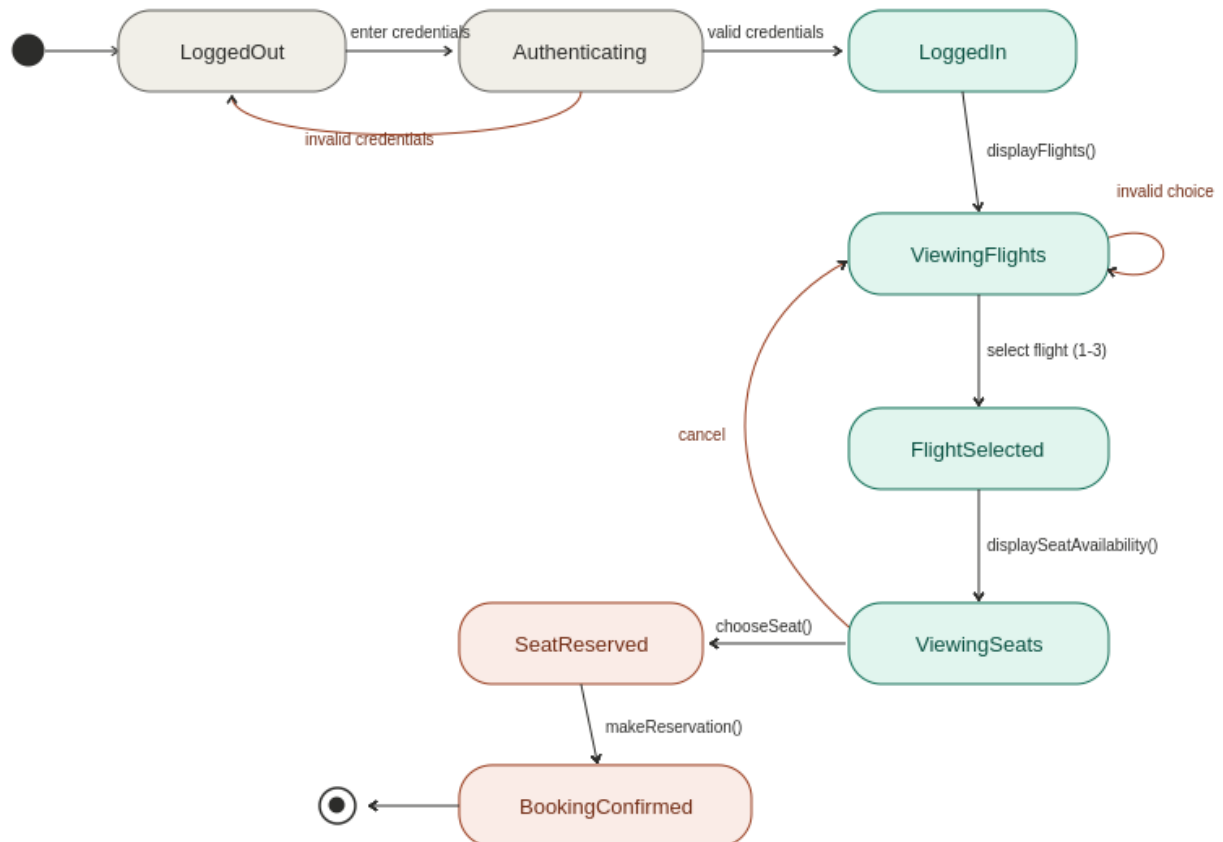


Explanation

- The outer alt frame branches on login validity; an invalid login returns an error and ends the interaction early.
- The inner alt frame branches on flight-selection validity; an out-of-range choice returns "Invalid choice" without contacting Flight or Reservation.
- Only the valid path reaches makeReservation(), which triggers Reservation.confirmBooking() and returns a reservation ID back up to the passenger.

State Machine Diagram

The state machine tracks a passenger's session as it moves through the booking process, from logged out through authentication, flight and seat selection, and finally to a confirmed booking. Invalid input at any decision point returns to an earlier state rather than advancing.



Key transitions

- LoggedOut to Authenticating to LoggedIn: driven by authenticateUser(); invalid credentials send the session back to LoggedOut.
- ViewingFlights loops on itself for an invalid selection, and only advances to FlightSelected once the choice is between 1 and 3.
- SeatReserved to BookingConfirmed is the terminal transition, corresponding to makeReservation() completing successfully.